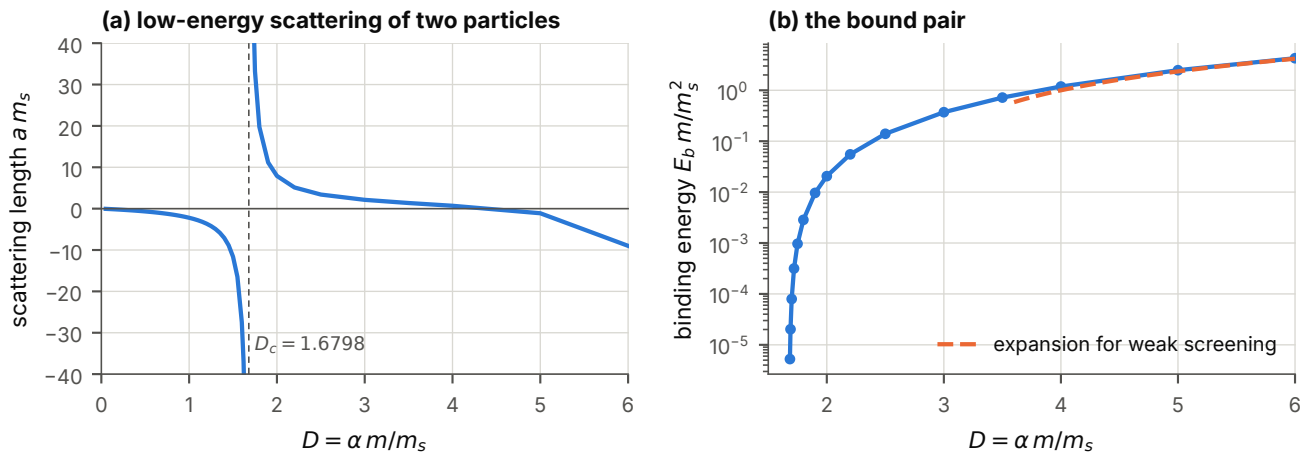
(a) low-energy scattering of two particles
scattering length $a\,m_s$
40
30
20
10
0
−10
−20
−30
−40
$D_c = 1.6798$
0
1
2
3
4
5
6
$D = \alpha\, m/m_s$
(b) the bound pair
binding energy $E_b\, m/m_s^2$
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
expansion for weak screening
2
3
4
5
6
$D = \alpha\, m/m_s$